Token usage run report: one Claude Code session

Run: session `784f9132-33b2-545c-8e00-c83cc57517f6`, read 2026-10-20 at the session's state then (UTC start 23:13). **Commit:** `748420928` (base of the change). **Instrument:** `tools/token_usage.py session` at the PR head. **Prices:** `tools/token_prices.json`, list prices as of 2026-10-06. All values are **measured** token counts from the transcript. Costs are **derived** from those counts × list price per request.

What ran: the core-tier main session (Opus 5.5) that wrote this change, and one Haiku 5.5 `Explore` subagent dispatched once as a probe of how subagent usage is recorded. 39 API responses in all.

agent type	model	requests	prompt tok	cache hit	output tok	cost (USD)	output cost share
main	opus-5-5	37	5058860	96.9%	38310	$2.991	25.6%
Explore	haiku-5-5	2	76335	49%	232 (≥)	$0.0054	2.2%
total		**39**	**5135195**	**96.2%**	**38542**	**$2.9964**	**25.6%**

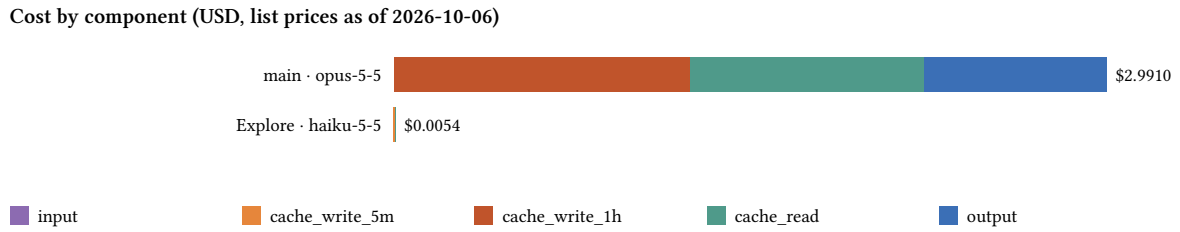

Figure 1: Cost by component per agent group, session 784f9132-33b2-545c-8e00-c83cc57517f6. One run, so no spread is recorded.

Derived values

- Uncached counterfactual (every prompt token at base input): $21.0094. Caching saving ratio 85.7% = 1 − $2.9964 / $21.0094.
- Share of output produced below the reasoning tiers: 0.6%. Share of cost: 0.2%.

Observations

- Summing transcript lines instead of responses overstated output 2.17× and prompt 1.94× at 65 lines / 34 responses (measured mid-session with the same instrument).
- The subagent's recorded output for its last response was 3 tokens. The Agent tool result's billed usage said 212. The first response's 20 tokens stay a lower bound.
- The subagent's first request wrote 37,408 tokens to cache (its system prompt and tools), so its cache hit ratio is 49% over two requests.

Limits

One session, one subagent dispatch: descriptive, no comparison. List prices, not invoices. Records: `record.json` beside this report.